Lesson 11.1 - Hypothesis Testing - Difference of Two Means

> **Video:** Hypothesis Testing - Difference of Two Means - Student's t-Distribution & Normal Distribution by The Organic Chemistry Tutor **Chapter:** 11 - Hypothesis Testing - 2 Samples

The big picture: from one sample to two

In Chapter 10 we compared **one** sample with a number that H_0 claimed (for example $\mu = 100$ g). In this chapter there is no outside number. We compare **two groups with each other**: two teachers, two factories, two batches, before vs after.

1. The question: is the gap real, or just luck? Even if two teachers were exactly equally good, their class averages would still differ a little, because each class is a random sample of students. The test asks whether the gap we see is bigger than chance alone would usually produce.

2. H_0 says "no difference". H_0 : $\mu_1 - \mu_2 = 0$. This gives H_0 the one exact number it needs (Lesson 10.1): the difference is **0**. H_a says the difference is not 0 (or is < 0, or > 0).

3. Turn two samples into one number: the difference. Compute $\bar{x}_1 - \bar{x}_2$, then use the same recipe as Lesson 10.4:

$$\text{test statistic} = \frac{\text{what we got} - \text{what } H_0 \text{ claims}}{\text{standard error}} = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE_{\text{difference}}}$$

4. Why the standard error gets bigger. Each sample mean has its own chance error. The difference picks up **both** of them, so it wobbles more than either mean alone. Variances add (never subtract):

$$SE_{\text{difference}} = \sqrt{SE_1^2 + SE_2^2} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

In Example 1 below, $SE_1 = 0.62$ and $SE_2 = 0.49$, so $SE_{\text{difference}} = \sqrt{0.62^2 + 0.49^2} = 0.79$. That is bigger than either one, but smaller than simply adding them (1.11).

Where the formula comes from. It follows from three rules about variance, where $\operatorname{Var}(X) = E[(X - \mu)^2]$ is the square of the SD.

Rule 1: scaling a variable scales its variance by the square.

$$\operatorname{Var}(cX) = E[(cX - c\mu)^2] = c^2\, E[(X - \mu)^2] = c^2 \operatorname{Var}(X)$$

So $\operatorname{Var}(-Y) = (-1)^2 \operatorname{Var}(Y) = \operatorname{Var}(Y)$. Flipping the sign doesn't shrink the spread.

Rule 2: for independent variables, variances add. Expand the square:

$$\begin{aligned}\operatorname{Var}(X + Y) &= E\big[\big((X - \mu_X) + (Y - \mu_Y)\big)^2\big] \\ &= \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\underbrace{E[(X - \mu_X)(Y - \mu_Y)]}_{\operatorname{Cov}(X,\,Y)}\end{aligned}$$

If X and Y are independent, $\operatorname{Cov}(X, Y) = 0$, so $\operatorname{Var}(X + Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)$.

Rule 3: so does the variance of a difference. Write $X - Y = X + (-Y)$ and use Rules 1 and 2:

$$\operatorname{Var}(X - Y) = \operatorname{Var}(X) + \operatorname{Var}(-Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)$$

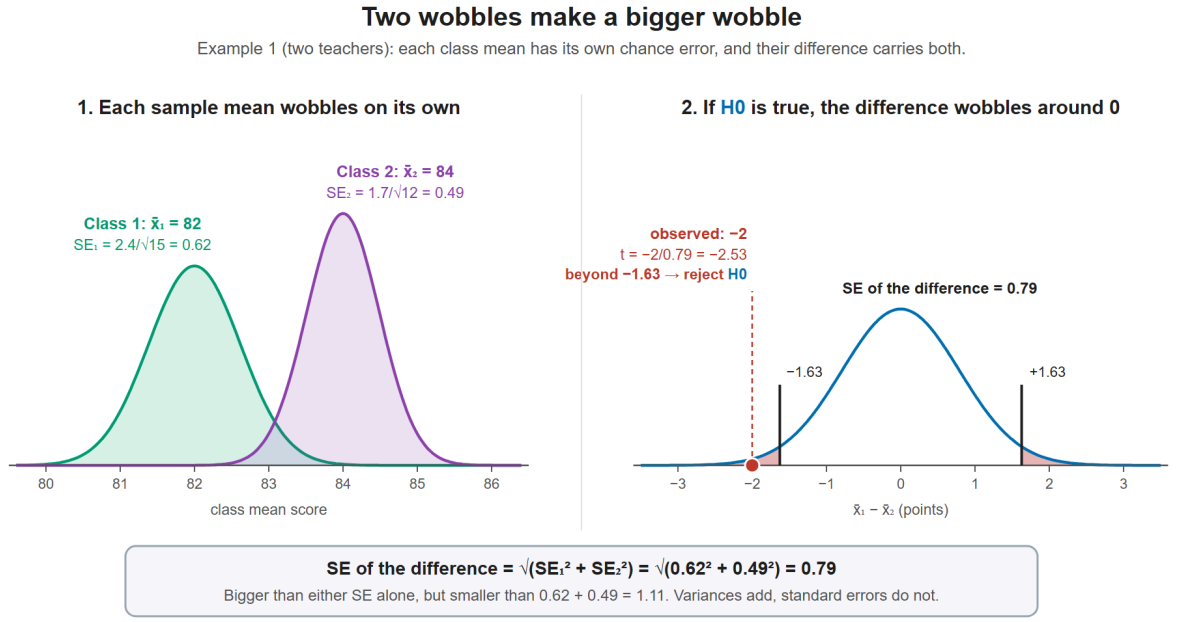
Now apply the rules to sample means.

- **One mean.** $\bar{x} = (X_1 + \cdots + X_n)/n$, with independent X_i that each have variance σ^2. Rule 2 gives $\mathrm{Var}(X_1 + \cdots + X_n) = n\sigma^2$. Rule 1 with $c = 1/n$ then gives $\mathrm{Var}(\bar{x}) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}$, so $SE = \sigma/\sqrt{n}$.
- **Two means.** The samples are independent, so Rule 3 gives:

$$\mathrm{Var}(\bar{x}_1 - \bar{x}_2) = \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2} \quad \Rightarrow \quad SE_{\text{difference}} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{SE_1^2 + SE_2^2}$$

We don't know σ_1 and σ_2, so we plug in s_1 and s_2. That's why the test uses t.

- **Like a right triangle.** Variances (squares) add, not SDs, so the SEs combine like the two shorter sides of a right triangle: $\sqrt{0.62^2 + 0.49^2} = 0.79$ is the long side. That's why it is bigger than each SE but less than their sum.
- **When the groups are not independent** (the same people measured twice), the covariance term stays: $\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y)$. Before and after weights move together, so the covariance is large and positive, and subtracting it removes most of the variance. That's why pairing helps (Lesson 11.4).



Left: each class mean has its own wobble. Right: if H_0 is true, the difference $\bar{x}_1 - \bar{x}_2$ wobbles around 0, and the wobble is wider than either one alone. The observed difference, -2, is 2.53 standard errors below 0, beyond the critical value, so we reject H_0.

5. "Independent" matters. This lesson assumes two separate groups of different people, with nothing linking someone in group 1 to someone in group 2. If the **same** people are measured twice (before and after), the groups are paired. Use Lesson 11.4 instead.

Which lesson for which question (Chapter 11 map):

What are you comparing?	Lesson	Tool
Means of two **independent** groups	11.1 (this one)	t (Welch) or z
How big is the difference between two means?	11.2	Cohen's d
Proportions (yes/no data) of two groups	11.3	z with a pooled $\bar{p}$
The **same subjects twice**, or matched pairs	11.4	paired t on the differences

The 5 steps are the same as in Chapter 10. Only the formula in step 4 changes:

1. Write H_0 (difference $= 0$) and H_a.
2. Choose α.
3. Use the sign in H_a to find the critical value(s).
4. **Compute the test statistic from both samples.** ← this lesson
5. Compare: beyond the critical value → reject H_0. Otherwise → fail to reject H_0.

Setup

Two **independent** groups. Does their population mean differ?

$$H_0 : \mu_1 = \mu_2 \;\; (\mu_1 - \mu_2 = 0) \qquad H_a : \mu_1 \neq \mu_2 \;\; (\mu_1 - \mu_2 \neq 0)$$

Test statistic

$$t \text{ (or } z) = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}}$$

Under H_0, $\mu_1 - \mu_2 = 0$. The denominator is the **standard error of the difference**.

Sample sizes	Use
Small (< 30)	**t-distribution** (Welch), with the degrees of freedom below
Large (≥ 30)	**z** (normal distribution)

Degrees of freedom (Welch-Satterthwaite):

$$\text{df} = \frac{\left(\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}\right)^2}{\dfrac{\left(s_1^2/n_1\right)^2}{n_1 - 1} + \dfrac{\left(s_2^2/n_2\right)^2}{n_2 - 1}}$$

Example 1 - two teachers (t-test)

Do two teachers' classes score differently? $\alpha = 0.05$.

	Class 1	Class 2
n	15	12
$\bar{x}$	82	84
s	2.4	1.7

Hypotheses: $H_0 : \mu_1 - \mu_2 = 0$, $H_a : \mu_1 - \mu_2 \neq 0$ (two-tailed, $\alpha/2 = 0.025$).

Both samples are small → t-test.

Degrees of freedom:

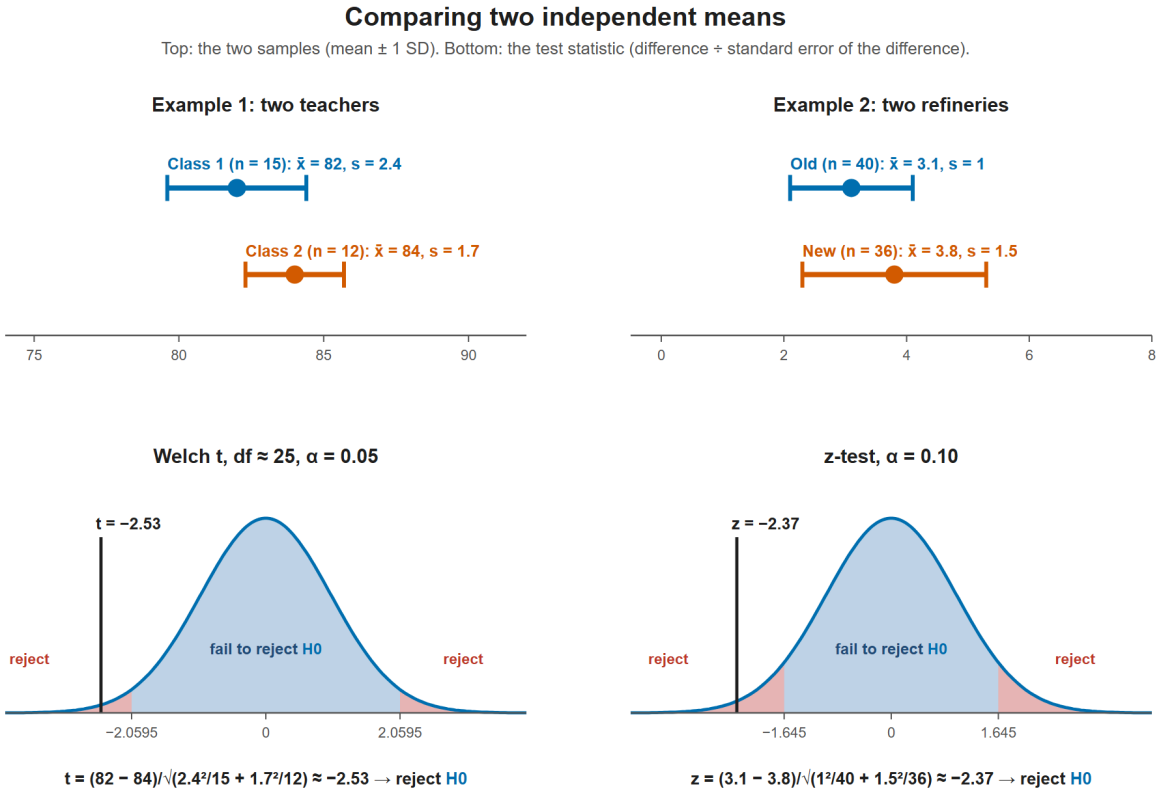
- $s_1^2/n_1 = 5.76/15 = 0.384$ and $s_2^2/n_2 = 2.89/12 = 0.24083$
- Numerator: $(0.384 + 0.24083)^2 = (0.62483)^2 = 0.390417$
- Denominator: $\dfrac{0.384^2}{14} + \dfrac{0.24083^2}{11} = 0.010533 + 0.005273 = 0.015805$
- $\text{df} = \dfrac{\text{numerator}}{\text{denominator}} = 0.390417/0.015805 \approx 24.7$

The video rounds this to **25**, giving $t_{0.025,\,25} = 2.0595$. Many textbooks round df **down** to be conservative ($24 \to t = 2.064$). Both give the same decision here.

Test statistic:

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\dfrac{s_1^2}{n_1} + \dfrac{s_2^2}{n_2}}} = \frac{82 - 84 - 0}{\sqrt{0.384 + 0.24083}} = \frac{-2}{0.79046} \approx -2.53$$

$-2.53 < -2.0595$, which is in the rejection region. **Reject H_0.** At the 5% level there is a significant difference between the two teachers' class means.



Top: the two samples in each example. Bottom: the difference in means divided by its standard error. In both examples the statistic lands in the left rejection region, so the means differ significantly.

Example 2 - two oil refineries (z-test)

Is there a difference in refining rate (liters per second)? $\alpha = 0.10$.

	Old factory	New factory
n	40	36
$\bar{x}$	3.1	3.8
s	1.0	1.5

Hypotheses: $H_0 : \mu_1 = \mu_2$, $H_a : \mu_1 \neq \mu_2$ (two-tailed).

Both samples are large → z-test. The confidence level is 90%, so the left area for the critical value is 0.95: $z = \pm 1.645$.

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{3.1 - 3.8}{\sqrt{\frac{1^2}{40} + \frac{1.5^2}{36}}} = \frac{-0.7}{\sqrt{0.025 + 0.0625}} = \frac{-0.7}{0.2958} \approx -2.37$$

$-2.37 < -1.645$ → **reject** H_0. At the 10% level there is a significant difference between the factories; the new one refines faster.

Key takeaways

- H_0: equal means (difference 0). Test statistic = (difference in sample means) ÷ $\sqrt{s_1^2/n_1 + s_2^2/n_2}$.
- The standard error of a difference is $\sqrt{SE_1^2 + SE_2^2}$: two sources of chance error, so it is bigger than either one alone.
- Small samples → t with Welch's df. Large samples → z.
- Compare with the critical value(s), or with the p-value, as in the one-sample case.